European Commission

PVGIS: Photovoltaic Geographic Information System

Performance of grid-connected PV

System Data

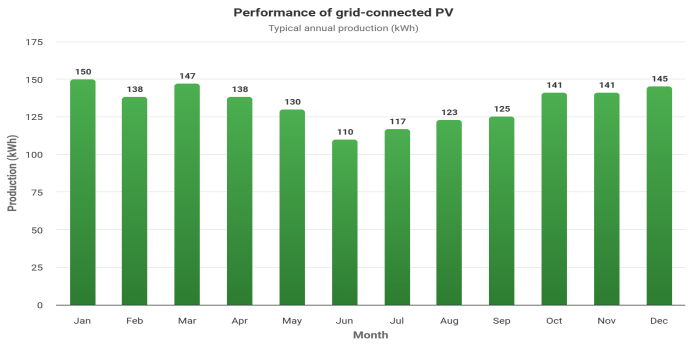
Parameter	Value
Optimal slope	8 °
Latitude	5.6037
Longitude	-0.1870
PV Technology	crystallineSilicon
Installed Power	1 kWp
System Loss	14 %
Mounting Position	freeStanding
Slope	8 °
Azimuth	180 °

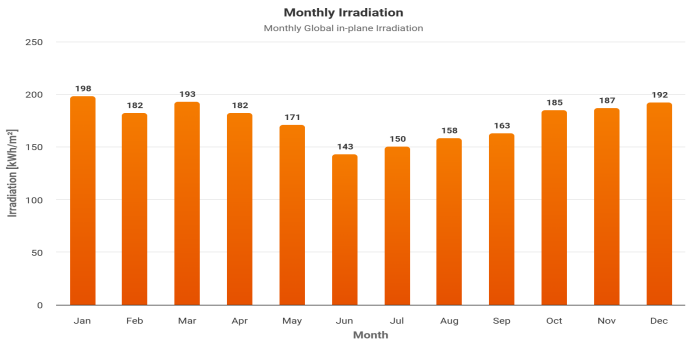
Simulation outputs

Output	Value
Yearly PV energy production [kWh]	1604
Year-to-year variability [kWh]	30
Yearly in-plane irradiation [kWh/m²]	2105

Factors impacting performance:

Effect	Value
Angle of incidence [%]	-1.63 %
Spectral effects [%]	0.38 %
Temperature and low irradiance [%]	-10.26 %
Total loss (including system losses) [%]	-23.79 %





Monthly PV energy and solar irradiation

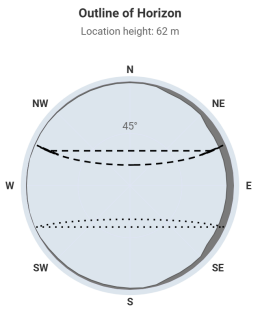
Month	E_m [kWh]	H(i)_m [kWh/m²]	SD_m [kWh]
Jan	149.6	198.2	0.00
Feb	137.7	182.0	0.00
Mar	146.6	192.8	0.00
Apr	138.3	182.5	0.00
May	130.0	171.3	0.00
Jun	109.8	142.7	0.00
Jul	116.6	150.3	0.00
Aug	122.8	157.9	0.00
Sep	125.5	162.6	0.00
Oct	141.0	185.0	0.00
Nov	141.3	187.2	0.00
Dec	144.7	192.2	0.00

E_m: Average monthly electricity production from the defined system [kWh].

H(i)_m: Average monthly sum of global irradiation per square meter received by the modules of the given system [kWh/m²].

SD_m: Standard deviation of the monthly electricity production due to year-to-year variation [kWh].

Outline of Horizon



Horizon height

Sun height, June (dashed line)

Sun height, December (dotted line)

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